BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, BAHAWALPUR

INTERMEDIATE (11TH CLASS) EXAMINATION, 2021

MATHEMATICS

Past Paper 2021

Retyped faithfully from the original board paper — same questions, no changes.

Paper I (Objective Type) — Inter-A-2021

پرچہ I (معروضی)

Time Allowed: 30 Minutes | **Maximum Marks:** 20 | **Paper Code:** 6193

Note: Four possible choices A, B, C, D to each question are given. Which choice is correct fill that circle in front of that Question No. Use Marker or Pen to fill the circles. Cutting or filling two or more circles will result in Zero Mark in that Question.

QQ.No.1.

Q1. If Cos x = -√3/2, then its Reference Angle is:

(A) π/3 (B) π/4

(C) π/6 (D) -π/6

Q2. Cos(Sec⁻¹(1)) =:

(A) 1 (B) 0

(C) 30° (D) 2

Q3. √(s(s-a)/bc) =:

(A) Sin α/2 (B) Sin β/2

(C) Cos β/2 (D) Cos α/2

Q4. If ABC be any Triangle and γ = 90°, then:

(A) $a^2 + b^2 = c^2$ (B) $a^2 + c^2 = b^2$

(C) $b^2 + c^2 = a^2$ (D) $a^2 + b^2 + c^2 = 0$

Q5. Period of Sin x/5 is:

(A) 2π (B) π/5

(C) 10π (D) 5π

Q6. 2 Sin12° Sin46° =:

(A) Cos34° Cos58°
(B) Sin34° + Sin58°
(C) Sin34° - Sin58°
(D) Cos34° - Cos58°

Q7. 3π/2 Radians equals to:

(A) 120°
(B) 150°
(C) 270°
(D) 190°

Q8. The Vertex of an angle in standard form is at:

(A) (0,0)
(B) (1,0)
(C) (0,1)
(D) (1,1)

Q9. In the Expansion of $(a+b)^7$, the 2nd term is:

(A) a^7
(B) $7a^6b$
(C) $7ab^6$
(D) $7b^6$

Q10. ${}^nP_n$ = ------:

(A) n!
(B) (n+1)!
(C) 1
(D) (n-1)!

Q11. Harmonic Mean between x and y is:

(A) 2(x+y)/xy
(B) 2xy/(x+y)
(C) (x+y)/2xy
(D) (x+y)/2

Q12. The nth term of the sequence 1/3, 2/5, 3/7, 4/9 ------- is:

(A) n/(2n-1)
(B) n/(2n+1)
(C) n/(3n-1)
(D) n/(3n+1)

Q13. The next two terms of the sequence 1,3,7,15,31 ------- are:

(A) 112,288
(B) 122,144
(C) 102,188
(D) 63,127

Q14. Partial Fractions of 1/x(x+1) are = ---------:

(A) 1/(x-1) + 1/(x+1)
(B) 1/(x-1) - 1/(x+1)
(C) 1/x + 1/(x+1)
(D) 1/x - 1/(x+1)

Q15. If α,β are the roots of the equation $x^2-4x+5=0$, then αβ is equal to:

(A) 2
(B) 4
(C) 5
(D) -4

Q16. (a+b)x = ax + bx is called:

(A) Identity
(B) Equation
(C) Conditional
(D) Fraction

Q17. If A = [[1,-2,3],[-2,3,1],[4,-3,2]], then A_{33} equals:

(A) -1
(B) 1
(C) 7
(D) -7

Q18. [[K,0],[0,K]] is:

(A) Zero Matrix
(B) Non-Diagonal Matrix
(C) Identity Matrix
(D) Scalar Matrix

Q19. If A ⊆ B and B ⊆ A, then:

(A) A = ∅
(B) A = B
(C) B = ∅
(D) A ∩ B = ∅

Q20. The Multiplicative Inverse of Complex Number (0,1) is:

(A) (0,-1)
(B) (0,1)
(C) (-1,0)
(D) (0,0)

Paper I (Subjective) — Inter-A-2021

پرچہ I (انشائیہ)

Time Allowed: 2:30 Hours | **Maximum Marks:** 80

Note: Session (2017-19) to (2020-22). It is compulsory to attempt any (8-8) Parts each from Q.No. 2 and Q.No.3 while attempt any (9) Parts from Q.No.4. Attempt any (3) Questions from Part-II. Write same Question No. and its Part No. as given in the Question Paper. [CAVEAT: the source PDF in this archive copy has only 2 pages; the second page ends with 'P.T.O.' — Q.No.4 parts beyond (iv) and all of Part-II are absent from this archive copy.]

Note: Part - I

Note: 25 x 2 = 50

Note: Q.No.2

(i) Find the Sum and Product of the Complex Numbers (8,9) and (5,-6).

(ii) Separate into Real and Imaginary Parts (2-7i)/(4+5i) and write as Simple Complex Number.

(iii) For all Complex Numbers Z, show that Z^2 + $Zbar^2$ is a real number.

(iv) Convert the theorem $(A \cap B)' = A' \cup B'$ into logical form and prove by constructing the Truth Table.

(v) If G is a group under the operation * and $a,b \in G$, then solve the equation $a * x = b$

(vi) Write the Descriptive Form and Tabular Form of the Set $\{ x \mid x \in O \wedge 3 < x < 12 \}$

(vii) If $A = [[1,2],[a,b]]$ and $A^2 = [[0,0],[0,0]]$ find the values of a and b.

(viii) Find the Co-factors A_{12} and A_{22} if $A = [[1,-2,3],[-2,3,1],[4,-3,2]]$

(ix) Find the value of x if $\det[[3,1,x],[-1,3,4],[x,1,0]] = -30$ | (x) If α,β are the roots of $5x^2-x-2=0$ then form an equation whose roots are $3/\alpha$ and $3/\beta$

(xi) Find Three Cube Roots of Unity. | (xii) Solve the Equation $2x^4-32=0$

> Note: Q.No.3

(i) Write $(3x^2-4x-5)/((x-2)(x^2+7x+10))$ in form of Partial Fraction without finding the constants.

(ii) Write $x^2/((x-2)(x-1)^2)$ in form of Partial Fractions without finding the constants.

(iii) Calculate $(2.02)^4$ by means of Binomial Theorem.

(iv) A die is rolled. What is the Probability that dots on the Top are greater than '4'?

(v) Use Binomial Theorem to expand $(x/2 - 2/x^3)^6$

(vi) Expand (4-3x)^(1/2) upto three terms taking the values of 'x' such that Expansion is valid.

(vii) Find a_8 of the sequence 1,-3,5,-7,9,-11,----- | (viii) Sum the Series $3/\sqrt{2} + 2\sqrt{2} + 5/\sqrt{2} +$ ----- $+ a_{13}$

(ix) Find two G.M.'s between 2 and 16. | (x) Which term of the A.P. 5,2,-1 ---- is -85?

(xi) Evaluate $20P_3$ | (xii) If ${}^nC_8 = {}^nC_{12}$ find 'n'

> Note: Q.No.4

(i) What is the circular measure of the angle between the hands of a watch at 4 O'Clock?

(ii) Verify $\cos2\theta = 2\cos^2\theta - 1$, when $\theta = 30°, 45°$

(iii) Prove that $\cos^4\theta - \sin^4\theta = \cos^2\theta - \sin^2\theta$ for all $\theta \in R$

(iv) Find the value of Cos 105°

> Note: [illegible in source scan: source PDF ends here with 'P.T.O.' — Q.No.4 parts (v) onwards and all of Part-II are absent from this archive copy]